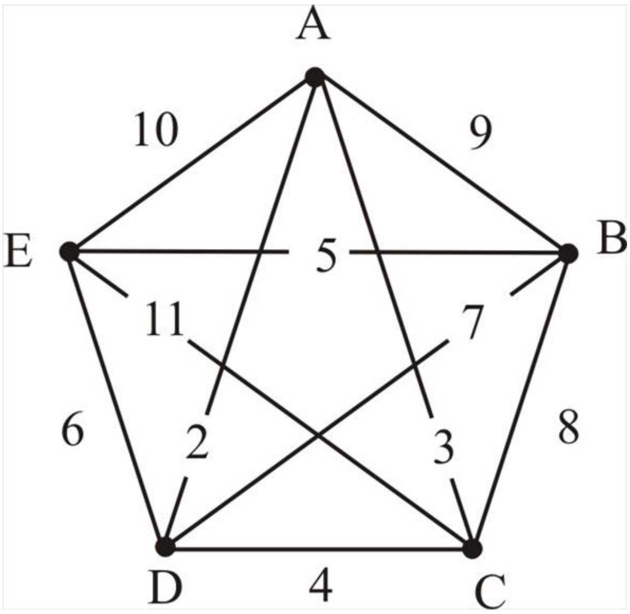
TIME COMPLEXITY OF MST ALGORITHMS **24/02/26**

1. Consider the following network.



   a. Use Prim's algorithm (starting at A) to find a minimum spanning tree for this network.
   b. How many comparisons were needed to determine which edge to add to the tree at the first step?
   c. After the first step, we have A and D. How many comparisons were needed to determine which edge to add to the tree next?
   d. How many comparisons were needed in total to find the MST?

2. Now consider the complete graph K_n. Find a formula for the number of comparisons needed to find an MST using Prim's algorithm. What is the Big O complexity of Prim's algorithm?